

Homework 6

Arham Lodha

February 27, 2024

Problem 1a.

Proof. For $\epsilon > 0$, by the Archimedean Property, $\exists K \in \mathbb{N}$ such that $K > \frac{\sqrt{5}}{\sqrt{4\epsilon}}$ thus $\epsilon > \frac{5}{4K^2}$. Then for any $n \geq K$:

$$\begin{aligned} \left| \frac{n^2 - 1}{2n^2 + 3} - \frac{1}{2} \right| &= \left| \frac{n^2 - 1 - \frac{1}{2}(2n^2 + 3)}{2n^2 + 3} \right| = \left| \frac{\frac{-5}{2}}{2n^2 + 3} \right| = \left| -\frac{5}{4n^2 + 6} \right| \\ &= \frac{5}{4n^2 + 6} \text{ since } n > 0 \implies 4n^2 + 6 > 0 \text{ and } |-5| = 5 \\ &\leq \frac{5}{4n^2} \\ &\leq \frac{5}{4K^2} \text{ since } n \geq K \\ &< \epsilon \text{ since } \epsilon > \frac{5}{4K^2} \end{aligned}$$

Thus $\lim \left(\frac{n^2 - 1}{2n^2 + 3} \right) = 0$.

□

Problem 5b.

Proof. For $\epsilon > 0$, by the Archimedean Property, $\exists K \in \mathbb{N}$ such that $K > \frac{1}{\epsilon}$ thus $\epsilon > \frac{1}{K}$. Then for any $n \geq K$,

$$\begin{aligned} \left| \frac{(-1)^n n}{n^2 + 3} - 0 \right| &= \left| \frac{(-1)^n n}{n^2 + 3} \right| = |(-1)^n| \left| \frac{n}{n^2 + 3} \right| \\ &= \left| \frac{n}{n^2 + 3} \right| \\ &\leq \left| \frac{n}{n^2} \right| = \left| \frac{1}{n} \right| = \frac{1}{n} \\ &\leq \frac{1}{K} \text{ since } n \geq K \\ &< \epsilon \text{ since } \epsilon > \frac{1}{K} \end{aligned}$$

Thus $\lim \left(\frac{(-1)^n n}{n^2 + 3} \right) = 0$.

□

Problem 2.

Proof. Let (x_n) be a sequence where $\lim x_n = x > 0$. Let $\epsilon = \frac{x}{20}$. Thus since $x > 0$, $\epsilon < \frac{x}{2} < x$ and $-\frac{x}{2} < -\epsilon$ and $\epsilon < x$. By definition of limit, for this ϵ , $\exists K \in \mathbb{N} \ni \forall n \geq K$, $-\epsilon < x_n - x < \epsilon$. Thus since $-\frac{x}{2} < -\epsilon$ and $\epsilon < x$, $-\frac{x}{2} < x_n - x < x$. Thus $-\frac{x}{2} + x = \frac{x}{2} < x_n < x + x = 2x$, thus $\frac{x}{2} < x_n < 2x$. $\square$

Problem 3a.

Proof. Base case: $n = 1$, $1 < 2$. Induction Hypothesis: $\forall n \in \mathbb{N}$, $n < 2^n$ Inductive Step: By inductive hypothesis, $2n < 2^{n+1}$. Since $n \in \mathbb{N}$, $n \geq 1$ thus $n + 1 \leq n + n = 2n$. Thus $n + 1 \leq 2^{n+1}$.

Thus by induction, $\forall n \in \mathbb{N}$, $2^n > n$.

$\forall h \in \mathbb{R} \ni h > 0$, let $m = \lceil h \rceil$, thus $m \geq h$ and $m \in \mathbb{N}$. Thus $m < 2^m$, thus $h < 2^m$. Thus h is not an upperbound. Thus $x_n = 2^n$ has no upper bound and thus is unbounded. By the contrapositive of the theorem that convergent sequences are bounded, x_n is unbounded thus divergent, not convergent. $\square$

Problem 3b.

Proof. Let $a_n = \frac{1}{n}$, $b_n = \frac{1}{n^2}$, $c_n = 4$, $d_n = 2$. $\forall \epsilon > 0$, by Archimedean property, $\exists K, L \in \mathbb{N} \ni K > \frac{1}{\epsilon}$ and $L > \frac{1}{\sqrt{\epsilon}}$, thus $\frac{1}{L^2} < \epsilon < \frac{1}{K}$. Also let $M = 1$. Thus $\forall k \geq K, l \geq L, m \geq M$,

$$\begin{aligned} |a_n - 0| &= \left| \frac{1}{k} - 0 \right| = \frac{1}{k} \\ &\leq \frac{1}{K} \text{ since } k \leq K \\ &< \epsilon \text{ since } \frac{1}{K} < \epsilon \end{aligned}$$

$$\begin{aligned} |b_n - 0| &= \left| \frac{1}{l^2} - 0 \right| = \frac{1}{l^2} \\ &\leq \frac{1}{L^2} \text{ since } l \leq L \\ &< \epsilon \text{ since } \frac{1}{L^2} < \epsilon \end{aligned}$$

$$|c_n - 4| = |4 - 4| = 0 < \epsilon$$

Thus $\lim a_n = 0$, $\lim b_n = 0$, $\lim c_n = 4$.

Let $x_n = (2 + \frac{1}{n})^2 = 4 + \frac{4}{n} + \frac{1}{n^2} = c_n + 4a_n + b_n$

Since scalar multiple of a sequence which converges also converges and the limit of a scalar multiple of a sequence is equal to the scalar multiple of the limit of the sequence, $\lim 4a_n = 4 \lim a_n = 0$. Since the sum of sequences which converge and their limit is equal to the sum of the limits, x_n converges because c_n , $4a_n$, and b_n converge, and $\lim x_n = \lim c_n + \lim 4a_n + \lim b_n = 4 + 0 + 0 = 4$. $\square$

Problem 3c.

Proof. $\forall \epsilon > 0$, by the Archimedean property, $\exists K \in \mathbb{N} \ni K > \frac{2}{\epsilon^2}$ thus $\epsilon > \frac{2}{\sqrt{K}}$. Thus $\forall n \geq K$,

$$\begin{aligned}
 |x_n - 1| &= \left| \frac{\sqrt{n} - 1}{\sqrt{n} + 1} - 1 \right| = \left| \frac{\sqrt{n} - 1 - \sqrt{n} - 1}{\sqrt{n} + 1} \right| = \left| -\frac{2}{\sqrt{n} + 1} \right| \\
 &= \frac{2}{\sqrt{n} + 1} \\
 &< \frac{2}{\sqrt{n}} \text{ since } \sqrt{n} < \sqrt{n} + 1 \\
 &\leq \frac{2}{\sqrt{K}} \text{ since } K \leq n \\
 &< \epsilon \text{ since } \frac{1}{\sqrt{K}} < \epsilon
 \end{aligned}$$

Thus $\lim x_n = 1$ □

Problem 3d.

Proof. Let $x_n = (-1)^n n^2 \forall n > 0$, let $m = \lceil h \rceil$. If $m \bmod 2 = 0$, $m > 1$ thus $m^2 > m$ and $(-1)^m m^2 = m^2 > m \geq h$. Thus h is not an upper bound. Otherwise let $n := m + 1$, thus $n \bmod 2 = 1$ and $n > 1$, thus $n^2 > n$ and $(-1)^n n^2 = -n^2 < -n < -m \leq -h$. Thus h is not a lower bound. Thus x_n is not bounded above and thus is unbounded. Thus by the contrapositive of the theorem that convergent sequences are bounded, since x_n is unbounded it is not convergent and thus is divergent. □

Problem 3e.

Proof. Let $x_n = \frac{(-1)^n}{n+2}$. $\forall \epsilon > 0$, by the Archimedean property, $\exists K \in \mathbb{N} \ni K > \frac{1}{\epsilon}$, thus $\epsilon > \frac{1}{K}$. Thus $\forall n \geq K$,

$$\begin{aligned}
 |x_n - 0| &= \left| \frac{(-1)^n}{n+2} \right| = \frac{1}{n+2} \\
 &< \frac{1}{n} \text{ since } n < n+2 \\
 &\leq \frac{1}{K} \text{ since } n \geq K \\
 &< \epsilon \text{ since } 1/K < \epsilon
 \end{aligned}$$

Thus $\lim x_n = 0$ □